

DATA SCIENCE

21CSS303T · SRM Institute · Sixth Semester

PYQ MASTER SHEET

All Questions · All Answers · Cross-Paper Analysis

Papers Covered	Part A MCQs	Part B Qs	Part C Qs
4 Papers	80 MCQs	28 Questions	8 Questions

Papers Analysed

Code	Exam	Part A	Part B	Notable Feature
08JF6	July 2024	20 MCQs	6 Qs	Online learning, scatter s=, vert= traps
17MA6	May 2025	20 MCQs	6 Qs	Subplots, annotate(), pivot(), pairplot()
14JA6	July 2025	20 MCQs	6 Qs	BeautifulSoup, IQR outliers, df.head/sort_values
10DA6	Dec 2025	20 MCQs	6 Qs	NEW: ML topics — KNN, Logistic Reg, Overfitting, PCA

★ **CRITICAL:** Dec 2025 added 8 ML MCQs (Q13-20) + 3 ML Part B questions + ML-based Part C. If your upcoming exam follows this pattern, you must know supervised/unsupervised basics.

PART A — MCQ Master List (All 4 Papers)

Topic 1: NumPy

Q	Paper	Question	Answer
1	May25	Identity matrix in NumPy?	numpy.eye()
1	Jul25	Faster in NumPy than Python lists?	Vectorized operations
3	Jul25	Function to create NumPy array?	array()
7	May25	Join two arrays in NumPy?	numpy.concatenate()
20	Jul25	Create array of zeros?	np.zeros()
1	Dec25	Identity matrix in NumPy?	eye()
8	Jul24	np prefix for mean function?	np (NOT npm/ng/ngm)
2	Jul24	eye() does what?	Generates identity matrix

Topic 2: Pandas

Q	Paper	Question	Answer
4	May25	Best for labeled 1D data?	Series
2	Dec25	Best for labeled 1D data?	Series
1	Jul24	1D labeled object in Pandas?	Series
4	Jul25	Difference: Series vs DataFrame?	Series=1D, DataFrame=2D
6	Jul24	DataFrame accepts?	Both ndarray AND Series
3	May25	df.describe() output?	Summary statistics
5	May25	.drop() does what?	Drops entries/columns from DataFrame
10	Jul25	Change index of DataFrame?	reset_index() and set_index()
14	Jul25	Read CSV in Pandas?	pd.read_csv()
17	Jul25	Sort DataFrame by column?	df.sort_values()
18	Jul25	First few rows of DataFrame?	df.head()
19	Jul25	Remove duplicate rows?	drop_duplicates()
13	May25	Purpose of pivot()?	To create a pivot table
3	Jul24	Purpose of reindexing?	To reset the index of a dataframe
5	Jul24	Data Alignment means?	Matching data based on common indices

Topic 3: Missing Data, Merging & Wrangling

Q	Paper	Question	Answer
10	May25	Handle missing data in Pandas?	fillna() and dropna()
5	Jul25	Remove missing values?	dropna()

3	Dec25	Remove missing values from DataFrame?	dropna()
8	May25	Data wrangling primarily for?	Cleaning and transforming data
9	May25	Merges on common index values?	merge_on_index()
7	Jul25	Merge on common column?	merge()
5	Dec25	Merge on common column?	merge()
6	Dec25	Why handle missing data?	Prevents model bias and errors
7	Jul24	After acquiring data, what's next?	Data cleaning

Topic 4: Data Preprocessing & Transformation

Q	Paper	Question	Answer
11	May25	Group data into bins?	Binning
13	Jul24	Binning purpose?	Split into categories/groups
12	May25	Raw data → clean dataset called?	Data Transformation
11	Jul24	Data transformation involves?	Converting text data into numeric format
14	May25	Standardization refers to?	Scaling to mean 0 and variance 1
14	Jul24	Standardization aims to?	Make data distribution more uniform
15	Jul24	Outlier approach?	Detecting and removing them
7	Dec25	Detect outliers using?	boxplot()
8	Jul25	Detect outliers method?	Interquartile Range (IQR)
8	Dec25	Categorical strings before ML?	One-hot encoding
12	Jul24	Summarizing technique in stats?	Aggregation

Topic 5: Visualization — Matplotlib & Seaborn

Q	Paper	Question	Answer
15	May25	Plotting library in Python?	Matplotlib
11	Jul25	Most common viz library?	Matplotlib
16	May25	Box plot visualizes?	Distribution and outliers
13	Jul25	Box plot purpose?	Display spread and detect outliers
10	Dec25	Boxplot primarily used to?	Show distribution and detect outliers
18	Jul24	plt.boxplot() orientation param?	vert (NOT Vertical, NOT Orientation)
16	Jul24	plt.scatter() size parameter?	s (NOT sizes, NOT marker_size)
19	Jul24	Save plot to file?	plt.savefig()
11	Dec25	Save plot to file?	plt.savefig()
17	May25	subplots() use?	Multiple plots in one figure
19	May25	Annotate text in Matplotlib?	annotate()
20	May25	Frequency distribution plot?	Histogram
12	Jul25	Distribution of numerical variable?	Histogram
12	Dec25	Histogram bins too wide — fix?	bins parameter
17	Jul24	Seaborn heatmap function?	seaborn.heatmap() — NOT plot_heatmap
18	May25	Seaborn for pairwise relationships?	pairplot()
9	Dec25	pairplot() comes from?	Seaborn
20	Jul24	Best viz library?	Seaborn

Topic 6: Large Data, Web Scraping & APIs

Q	Paper	Question	Answer
6	Jul25	Main challenge of large datasets?	High computational cost and memory usage
9	Jul24	Online learning handles large data by?	Processing one observation at a time
10	Jul24	Advantage of online learning?	Trained without loading all data into memory
9	Jul25	Web scraping purpose?	Extract structured data from web pages automatically
16	Jul25	Library for web scraping?	BeautifulSoup
6	May25	NOT used for data acquisition?	Image rendering

Topic 7: Machine Learning (NEW — Dec 2025)

★ These questions ONLY appeared in Dec 2025. Watch for them if your exam follows that pattern.

Q	Paper	Question	Answer
13	Dec25	Supervised learning algorithm?	Linear Regression
14	Dec25	Purpose of train/test split?	Evaluate model on unseen data
15	Dec25	Categorical dependent variable?	Logistic Regression
16	Dec25	High train, low test accuracy?	Overfitting
17	Dec25	Clustering algorithm?	K-means
18	Dec25	Primary goal of unsupervised learning?	Find hidden patterns in data
19	Dec25	Dimensionality reduction method?	PCA
20	Dec25	Wrong cluster assignment cause?	Poor choice of number of clusters

PART B — All Questions Grouped by Theme

Theme 1: Data Science Process & NumPy/Pandas Foundations

Paper	Q No.	Question
May25	Q21	Key steps involved in Data Science Process with neat sketch
Jul25	Q21	NumPy: write code to calculate mean, max, min + sort temperatures array
Dec25	Q21	NumPy + Pandas for retail store data — array creation, indexing, slicing, joining, aggregation

Theme 2: Large Data Handling

Paper	Q No.	Question
May25	Q22	Techniques to handle large datasets + challenges faced during data wrangling
Jul25	Q27a	Challenges with large datasets — provide 3 general techniques (5 marks)
Jul25	Q27b	Data wrangling operations: merge on index, concatenate datasets, reshape using pivot (5 marks)
Jul25	Q27c	Fill missing Salary with median per dept; remove Age outliers using IQR method (5 marks)

Theme 3: Data Cleaning & Wrangling (HIGHEST FREQUENCY)

★ This theme dominates Part B AND Part C across all papers. Master this theme completely.

Paper	Q No.	Question
May25	Q23	Python code: clean + transform using dropna(), fillna(), replace()
May25	Q26	Handle missing data + common techniques to deal with missing values
Jul25	Q24	Step-by-step: identify duplicate records + missing values + standardization importance
Dec25	Q22	Healthcare data: missing values, inconsistent categories, extreme outliers — full wrangling plan
Dec25	Q27	COVID-19 data: load DataFrame, handle missing, standardize city names, remove age outliers
Dec25	Q26	Social media text pipeline: cleaning, tokenization, stopword removal, normalization

Theme 4: Visualization

Paper	Q No.	Question
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May25	Q24	Explain line plots, scatter plots, box plots, and histograms used in data visualization
May25	Q25	Types of visualizations + Matplotlib/Seaborn customization — 3 key techniques
Jul25	Q23	Scatter plot for house price vs size data + Seaborn regression line
Jul25	Q25	Types of visualizations + Matplotlib/Seaborn customization
Jul25	Q26	Seaborn: scatter (Marketing Spend vs Sales Revenue) + regression line
Jul24	Q25	Formatting: ticks, labels, legends in Matplotlib with examples
Jul24	Q26	3D surface plot using Python and Matplotlib — full code
Dec25	Q23	Financial data: line, bar, histogram, boxplot, pairplot — when and why to use each

Theme 5: Merging & Concatenating

Paper	Q No.	Question
May25	Q25	Steps in merging and concatenating datasets using Pandas
Jul25	Q27b	Merge on index + concatenate + reshape via pivot — with examples
Jul24	Q23	Merging on index labels — implications and use cases

Theme 6: Web Scraping & APIs

Paper	Q No.	Question
Jul25	Q22a	What is Web Scraping? Applications and ethical concerns (5 marks)
Jul25	Q22b	Compare Web APIs and Open Data Sources as data acquisition methods (5 marks)
Jul24	Q24	Web APIs: explain + provide Python code snippet to retrieve data

Theme 7: Machine Learning (NEW — Dec 2025)

★ Only appeared in Dec 2025. Prepare if your exam may follow this pattern.

Paper	Q No.	Question
Dec25	Q24	E-commerce: supervised learning — Logistic Regression/Decision Trees, train/test split, accuracy/precision/recall/F1
Dec25	Q25	Telecom: K-means clustering — preprocessing, number of clusters, silhouette score
Dec25	Q28	Insurance: Logistic Regression vs Decision Tree — encode + normalize + train + evaluate

PART C — All Questions (15 Marks Each)

Paper	Q27	Q28
July 2024	Data Science lifecycle for e-commerce company — data-driven insights	(i) Data wrangling techniques with Python (ii) String manipulation with examples
May 2025	3D surface plot using Matplotlib — full Python program with visualization techniques	Data cleaning and wrangling techniques on a sample dataset using Pandas
July 2025	Large data challenges (5m) + wrangling: merge on index / concat / pivot (5m) + missing/outlier code IQR (5m)	E-commerce sales: explain plots (5m) + Matplotlib 4 subplots (5m) + Matplotlib/Seaborn customization (5m)
Dec 2025	COVID-19 data: identify quality issues (5m) + full wrangling Python code (10m)	Insurance: Logistic Regression vs Decision Tree (5m) + full ML Python code (10m)

★ **DATA WRANGLING** appears in Part C of ALL 4 papers in some form. This is your guaranteed 15 marks.

Pattern Analysis & Exam Strategy

MCQ Topic Frequency

Topic	Papers (out of 4)	MCQs per paper	Priority
Pandas (Series, DataFrame, functions)	4/4	4-6	MUST KNOW
Visualization (Matplotlib + Seaborn)	4/4	4-8	MUST KNOW
Data Preprocessing (binning, std, outlier)	4/4	3-5	MUST KNOW
NumPy (eye, array, concatenate)	4/4	2-4	MUST KNOW
Missing data + Merging	4/4	2-4	HIGH
Large data + Online learning	3/4	1-3	HIGH
Web scraping + APIs	2/4	1-2	MEDIUM
ML (supervised, unsupervised, K-means)	1/4	8 (Dec25 only)	NEW — WATCH

Part B Frequency by Theme

Theme	Frequency	Safe to prepare
Data Wrangling / Cleaning	4/4 papers	Yes — guaranteed
Visualization	4/4 papers	Yes — 1-2 questions always appear
Large Data Handling	3/4 papers	Yes — very reliable
Merging / Concatenating	3/4 papers	Yes — often paired with wrangling
NumPy + Pandas foundations	2/4 papers	Safe Part B option
Web Scraping + APIs	2/4 papers	Prepare as 5th backup
Machine Learning	1/4 papers	Only if Dec 2025 pattern repeats

Recommended 4 Part B Questions to Prepare

#	Question	Why
1	Large data challenges + 3 techniques	Appeared in 3/4 papers. Easy 10 marks with 5 bullet points
2	Merging on index + concatenate + pivot	Appears every paper. Code is short and reusable
3	Missing data + outlier handling (IQR)	Core of every wrangling question. 3-4 functions = 10 marks
4	Visualization: line/scatter/box/histogram + customization	4/4 papers have this. Know what each chart is for

Part C Strategy

Choice	Strategy
Safest (all papers)	Data wrangling + cleaning: define wrangling → missing values (fillna/dropna) → merge → string manipulation → transform → groupby. Always appears.
If viz question appears	Choose it ONLY if you can write code for: 4 subplots + scatter + histogram + boxplot + plt.tight_layout() + customization
If ML question appears	Choose ONLY if you know: train_test_split + fit() + predict() + accuracy_score() + confusion_matrix()

Quick Answer Templates

Wrangling — Standard Answer Structure (Part C)

```
import pandas as pd
df = pd.read_csv('data.csv')
df.isnull().sum()                # find missing
df.fillna(df.mean(), inplace=True) # fill numeric
df.dropna(inplace=True)          # drop remaining
df.drop_duplicates(inplace=True)  # remove duplicates
df['col'] = df['col'].astype(float) # fix dtype
Q1 = df['col'].quantile(0.25)     # IQR outlier removal
Q3 = df['col'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['col'] >= Q1-1.5*IQR) & (df['col'] <= Q3+1.5*IQR)]
merged = pd.merge(df1, df2, on='id', how='inner') # merge
df['name'].str.lower().str.strip() # string cleaning
```

3D Surface Plot Template (May 2025 Part C)

```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
import numpy as np
fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
X, Y = np.meshgrid(np.linspace(-5,5,50), np.linspace(-5,5,50))
Z = np.sin(np.sqrt(X**2 + Y**2))
ax.plot_surface(X, Y, Z, cmap='viridis')
plt.show()
```

Aswath AS · SRM Institute

"I got bored studying. So I analysed all 4 previous year papers."

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